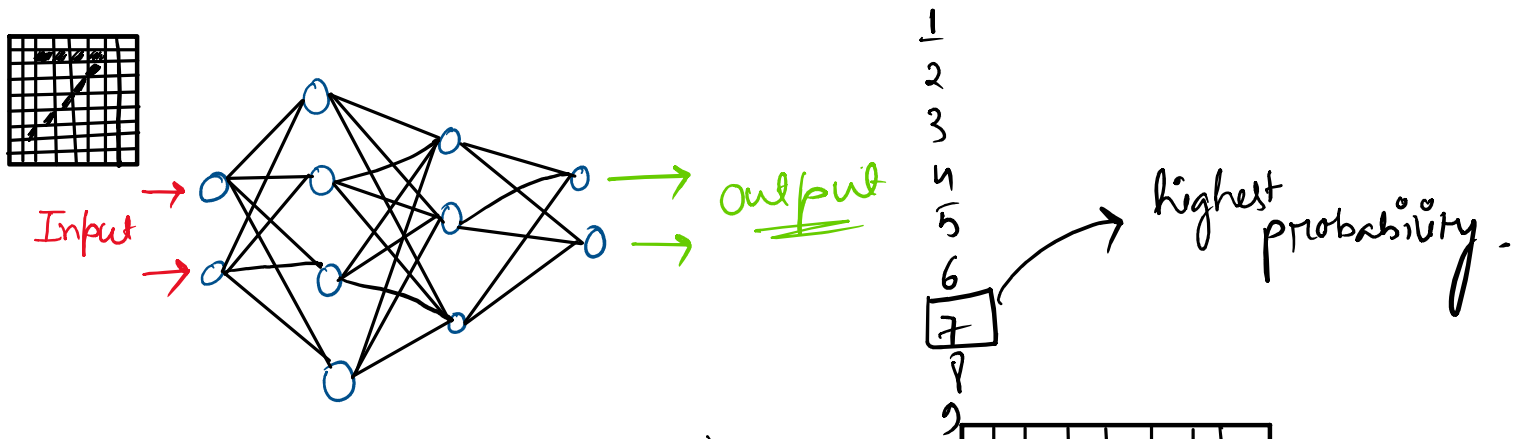


Neural Network → Not as a buzzword
but as a piece of Math.



Dataset: MNIST (60K images).
28x28.

Convolution Neural Network
→ Image recognition.

LSTM → Speech recognition
(sequence modeling).

Neural network are inspired from brain
human Neural system.
How they are connected?
What are neurons?

↳ + holds a number

→ $\textcircled{0.8}$ → Thing that holds a number.

Consider the image it has 28×28 dim.

So, $28 \times 28 = 784 \rightarrow$ neurons.

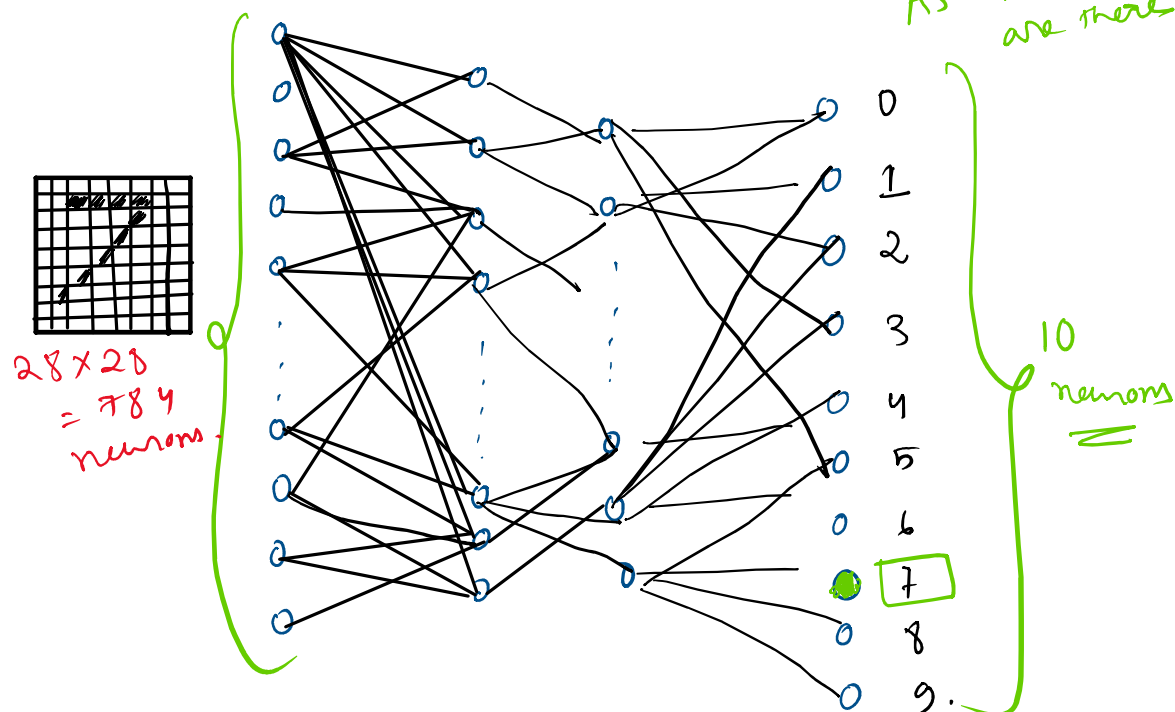
The value inside that neuron is

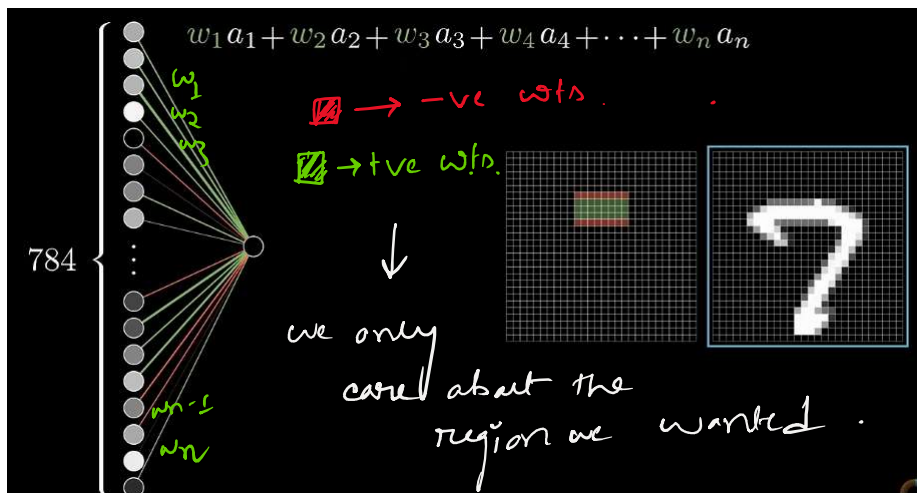
Activation. $\leftarrow \textcircled{0.8}$

What is neuron firing?

→ It's activation is a high number.

Design a Neural Network (MNIST).





Weighted sum = $w_1 a_1 + w_2 a_2 + \dots + w_n a_n$.

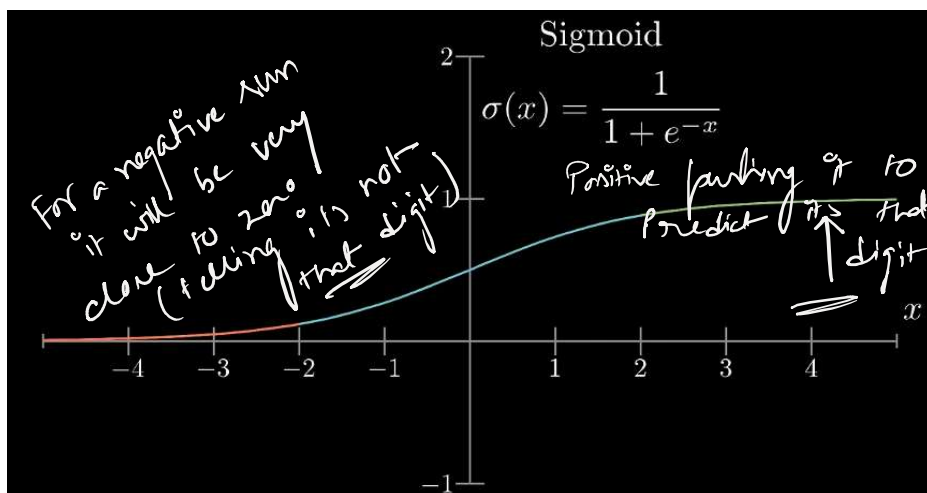
$\in (-\infty, +\infty)$

↓ but to send it to next neuron it should be between 0 and 1

$\Sigma \in [0, 1]$. squeezes between $[0, 1]$.

So, here comes sigmoid.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



$$G(w_1 a_1 + \dots + w_n a_n) \in [0, 1]$$

+b

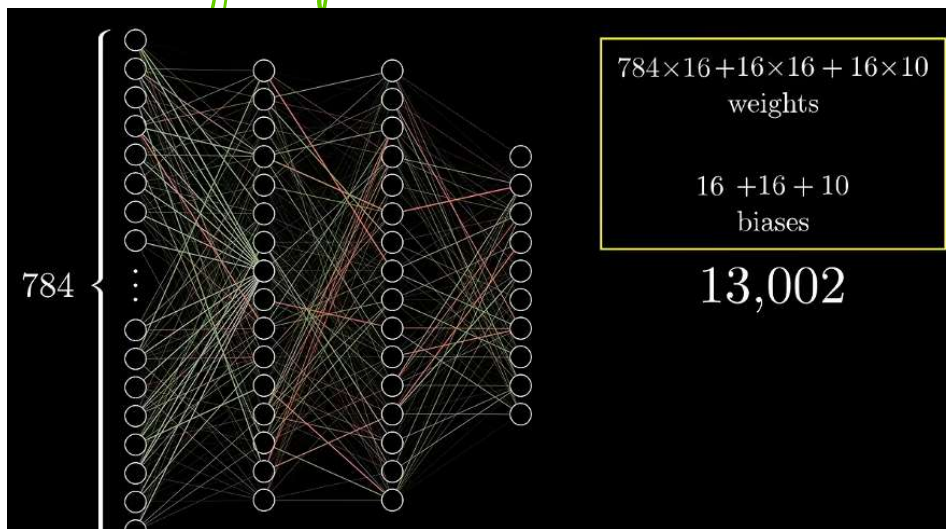
Only active meaningfully.

When $(w_1 a_1 + \dots + w_n a_n) > 10$

Bias for inactivity

bias (b).

↳ It tells you how the weighted sum must be to activate the neuron meaningfully.



1st Layer

No. of neurons = 784

No. of wt for 1 neuron = 16.

$$b = 16$$

$$W = 784 \times 16.$$

2nd Layer

No. of neurons = 16

No. of wt for 1 neuron = 16.

3rd Layer

$$16 \times 10$$

No. of neurons = 16

No. of wt for 1 neuron = 16.

16 x 10

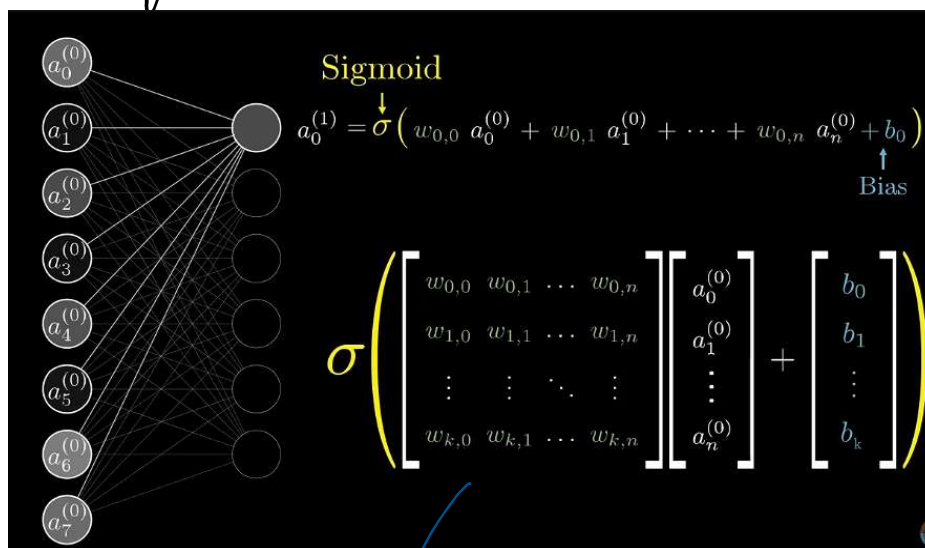
b = 10

w = 16 x 16, b = 16

Add up to 13,002.

So when we talk about Machine Learning

Finding the right weights and biases.



$$a^{(1)} = \sigma(Wa^{(0)} + b)$$

Remember at first I told that Neuron is just a number, but it's actually

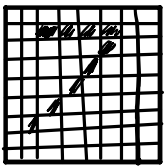
a function

$$f(a_0, \dots, a_{783}) = \begin{bmatrix} y_0 \\ \vdots \\ y_9 \end{bmatrix} \quad (\text{considering MNIST data})$$

$[y]$

Now a general question come

→ How does this network learn the appropriate weights and biases just by looking at the data?



→ 7 (It's actually 7)

Note: Instead of sigmoid now people use

ReLU → why?

$f(x) = \max(0, x)$ → easier to train

Motivation — ?

→ Our biology. (our neurons are either active or not.)

